

# 3.1

## BW Modeling Tools

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### OBJECTIVE

This exercise presents various tools, which are used in the SAP BW/4HANA 2.0 curriculum and provides the basic knowledge on when and how they can be applied.

### ACTIVITIES

Learn to use SAP BW modeling tools

### SOFTWARE PREREQUISITES

- Eclipse with SAP BW Modeling Tools
- SAP GUI

### DATA SET

Global Bike (GB) data set

### SYSTEM ACCESS

SAP BW/4HANA 2.0 from your UCC

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## Scenario

Jan Schmidt is Controlling Specialist at Global Bike, Germany. His manager, Hanan Cengiz, assigns him to explore the latest BW solution: SAP BW/4HANA 2.0. Jan's task is to create and implement a data model in the BW/4HANA 2.0 for Global Bike Germany and US with

the result that Hanan can represent their sales and cost in a faster way to the Chief Financial Officer (CFO) Wilhelm Ollenhauer. Jan wants to set up reporting to serve different needs.

## Introduction to Modeling tools using BW

1 - Modeling Tools

2 - Modeling Basics

3 - Master Data

4 - Data Acquisition

5 - Data Warehouse

6 - Hierarchies

7 - Slowly changing dimensions

The data warehouse is built in the SAP BW4/HANA 2.0 system. The interface for working with BW is Eclipse. Eclipse is an integrated development environment (IDE) used in computer programming and is the most widely used Java IDE<sup>1</sup>. It has several plug-ins for working with SAP tools such as BW Modeling Tools (BWMT).

Data Analytics is done by various tools like Microsoft Excel with the SAP Add-on SAP BusinessObjects Analysis for Office which can be used for multidimensional analysis. Dashboards that are built with Lumira Designer and typically executed in a Web Browser or a mobile device.

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<sup>1</sup> [https://en.wikipedia.org/wiki/Eclipse\\_\(software\)](https://en.wikipedia.org/wiki/Eclipse_(software))

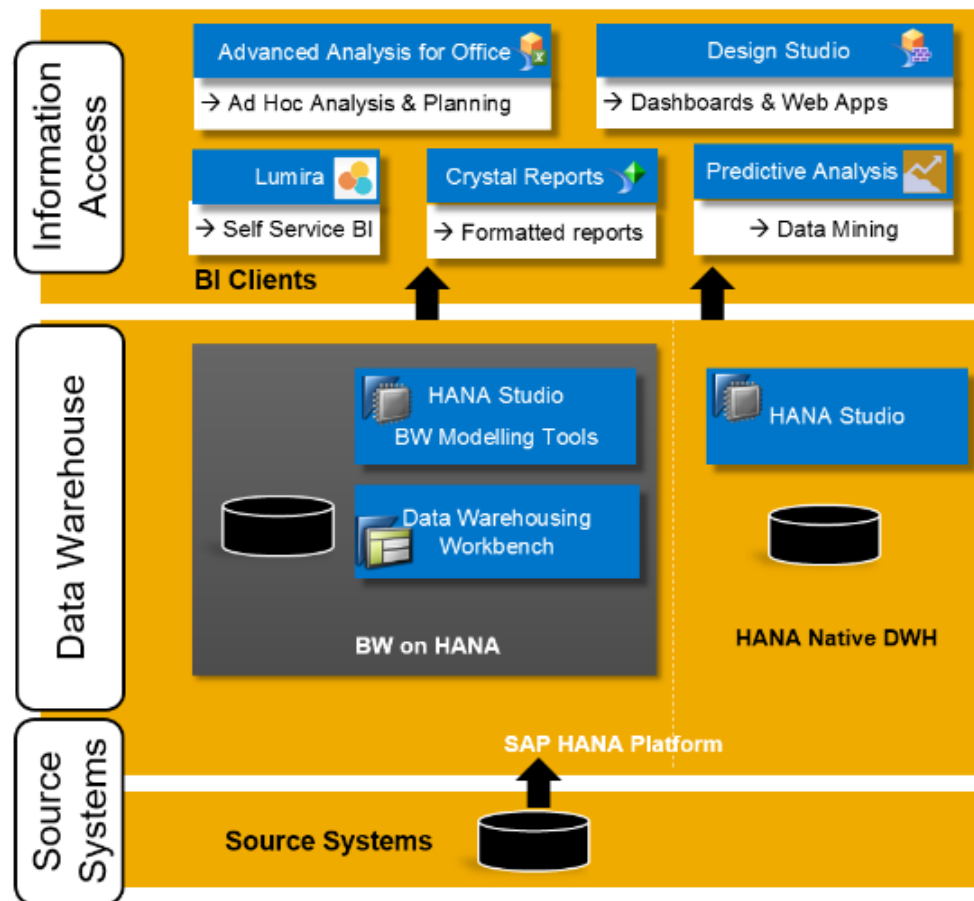


Figure 1

## General Notes and Caution

**Warning:** All exercises are performed in a central SAP BW4/HANA 2.0 system hosted by your UCC. This curriculum has been tested only on Windows. Since SAP BW/4HANA 2.0 system at your UCC has only one client, several universities are working in one single system. Therefore, all objects of different courses and participants have to be separated by naming conventions to differentiate their work. Each course has to be assigned a course ID, which is represented by a three-digit identifier XXX (e.g. AA1) and will be assigned to your class/instructor by your UCC. In addition, every student is assigned an additional three-digit identifier ### by the instructor. When working with the exercises, students have to replace XXX and ### in all technical names and descriptions with the numbers given to them by their instructor. There are some exceptions to this rule when students will use a UCC provided object. These will be explained in following exercises.

**Saving:** In this document, the instructions will tell you to save data and when it does, always click *enter* to validate your entries. If there are error(s) that occur, they must be corrected before continuing to the next task. You will find errors when there are red icons that appear in the message area. Messages that are yellow contain warnings and not errors. At times,

these must be fixed to move on. However, most of the time, they do not need to be fixed and you can simply click *enter* through them. Green messages indicate that nothing went wrong.

## 1. Introduction to Eclipse

The first-time starting Eclipse you must select a workspace. This will create a folder on your local system where all your Eclipse projects will be stored.

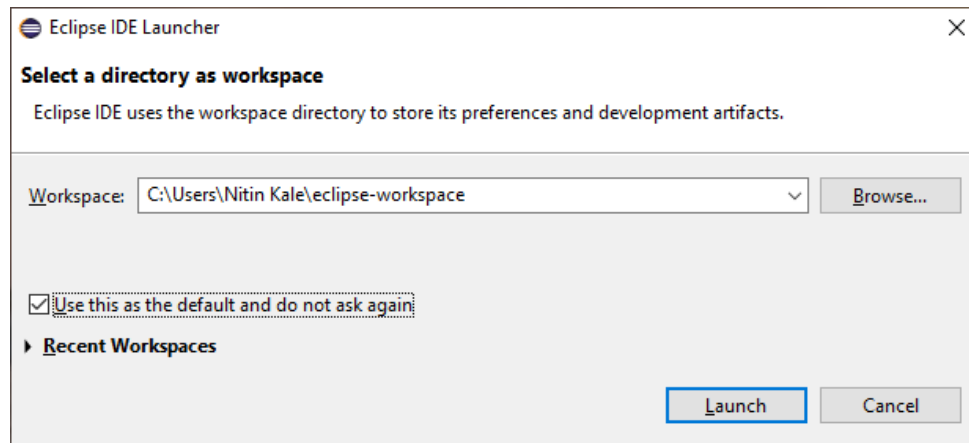


Figure 2

*Note: Choose a location on your computer per your convenience.*

- a. **Launch.**
- b. Open the BW Modeling Perspective. Go to **Window** → **Perspective** → **Open Perspective** → **Other** and select **BW Modeling**.

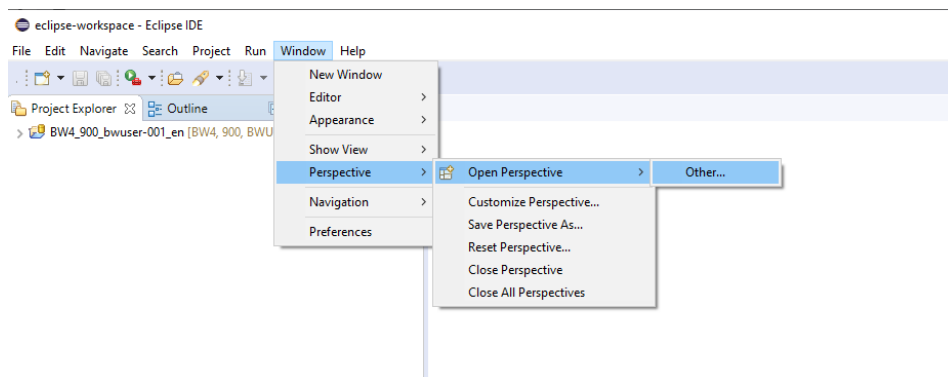


Figure 3

- c. The new perspective opens. If a Welcome Screen opens, close it first. Then you see your new perspective.

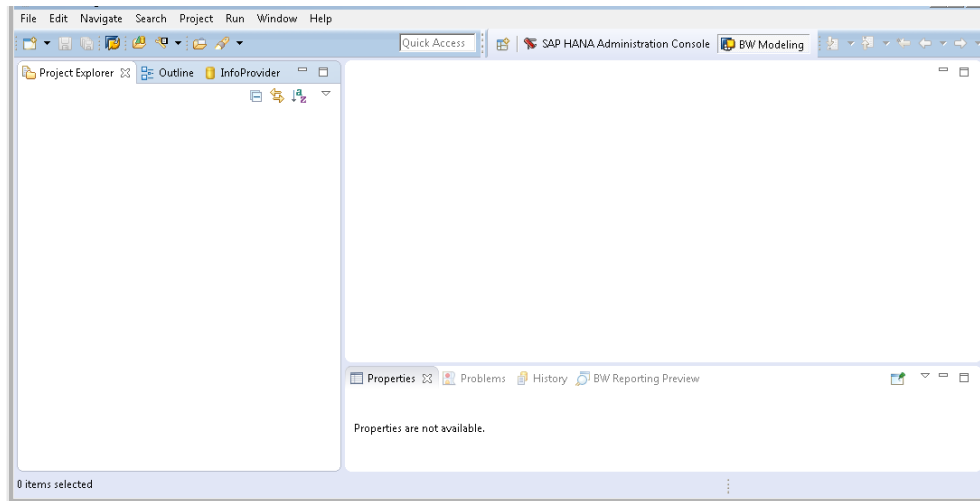


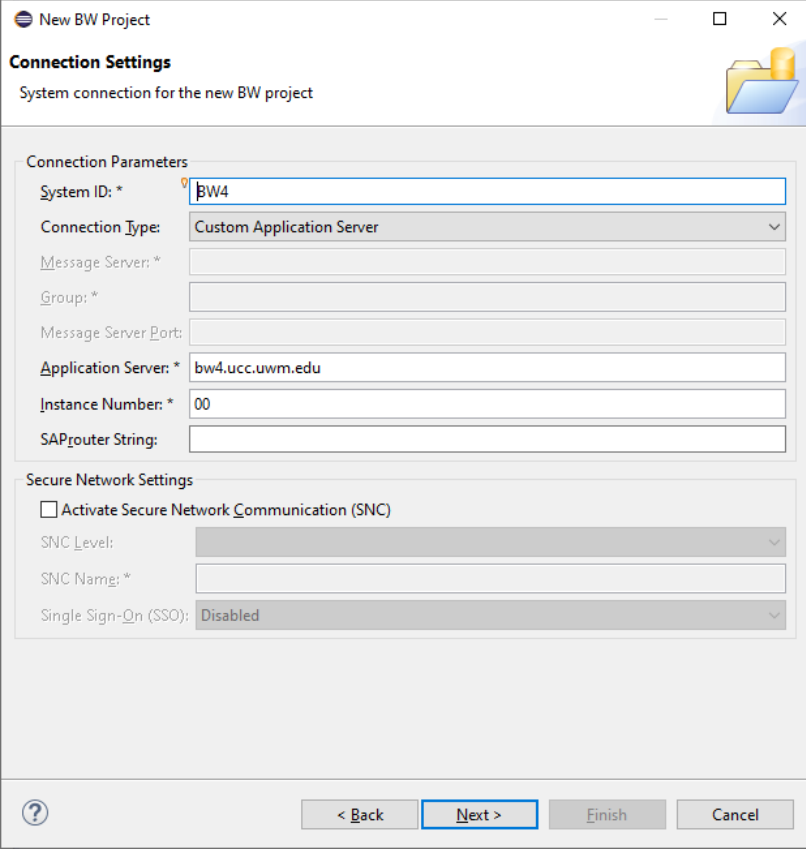
Figure 4

- d. Create a new BW Project. Go to **File** → **New** → **BW Project**.



Figure 5

- e. Select the system assigned by your instructor and click on **Next**.



**New BW Project**

**Connection Settings**  
System connection for the new BW project

**Connection Parameters**

System ID: \* BW4

Connection Type: Custom Application Server

Message Server: \*

Group: \*

Message Server Port:

Application Server: \* bw4.ucc.uwm.edu

Instance Number: \* 00

SAProuter String:

**Secure Network Settings**

☐ Activate Secure Network Communication (SNC)

SNC Level:

SNC Name: \*

Single Sign-On (SSO): Disabled

< Back **Next >** Finish Cancel

Figure 6

Note that the connection settings are the same as the connection settings in SAP GUI. If you face problems at this stage, investigate your connection settings in SAP GUI.

f. Then, click on **Next** again.

**New BW Project**

**Connection Settings**  
System connection for the new BW project

Connection Parameters:

System ID: \* BW4

Connection Type: Custom Application Server

Message Server: \*

Group: \*

Message Server Port:

Application Server: \* bw4.ucc.uwm.edu

Instance Number: \* 00

SAProuter String:

Secure Network Settings:

☐ Activate Secure Network Communication (SNC)

SNC Level:

SNC Name: \*

Single Sign-On (SSO): Disabled

< Back Next > Finish Cancel

Figure 7

- g. Click **Next**.
- h. Login to the system using the credentials (Client number, Username, Password) provided to you by your instructor and click on **Finish**.

**New BW Project**

**Login to System**

Specify a value for field 'Password'

System ID: BW4

Client: \* 900

User: \* AA1-199

Password: \*

< Back Next > Finish Cancel

Figure 8

*Note: The new BW project appears in the Project Explorer on the left-hand side. Look in the BW Repository for the InfoArea (folder) assigned by your instructor which has the technical name XXX### and description Student ###.*

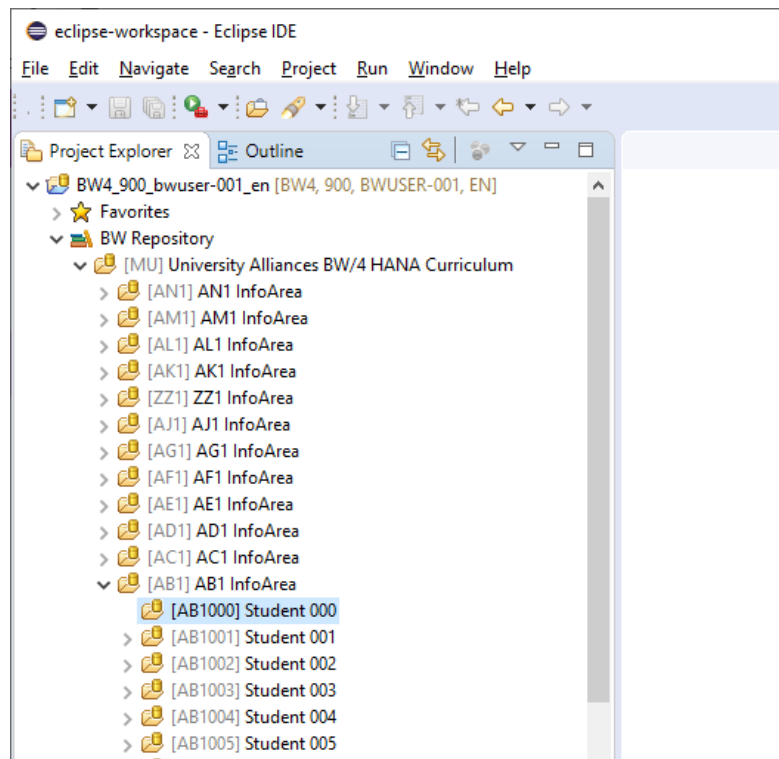


Figure 9

Your InfoArea is **AB1000** in this example and the course ID is **AB1**. Your student identifier is 000 in this example. Usually 000 is reserved for the instructor.

*DO NOT edit or delete any objects in the MU0 Master Copy InfoArea! DO NOT edit or delete any objects that don't belong to you!*

**Question 1:** Insert a screenshot of your student InfoArea (folder) for this semester. What is your InfoArea technical name? What is your InfoArea description?

**Question 2:** Are you permitted to edit or delete objects in the MU0 InfoArea?

**Question 3:** Are you permitted to edit or delete objects in InfoAreas that do not belong to you?

## 2. CompositeProvider

A CompositeProvider is an InfoProvider that combines data from various objects and/or from other InfoProviders(by union or inner join or left outer join), and makes this data available for analytics.

- a. Open Eclipse and go to your Info Area (e.g. AB1000) provided by your instructor. Right click on your **InfoArea** → **New** → **CompositeProvider**.

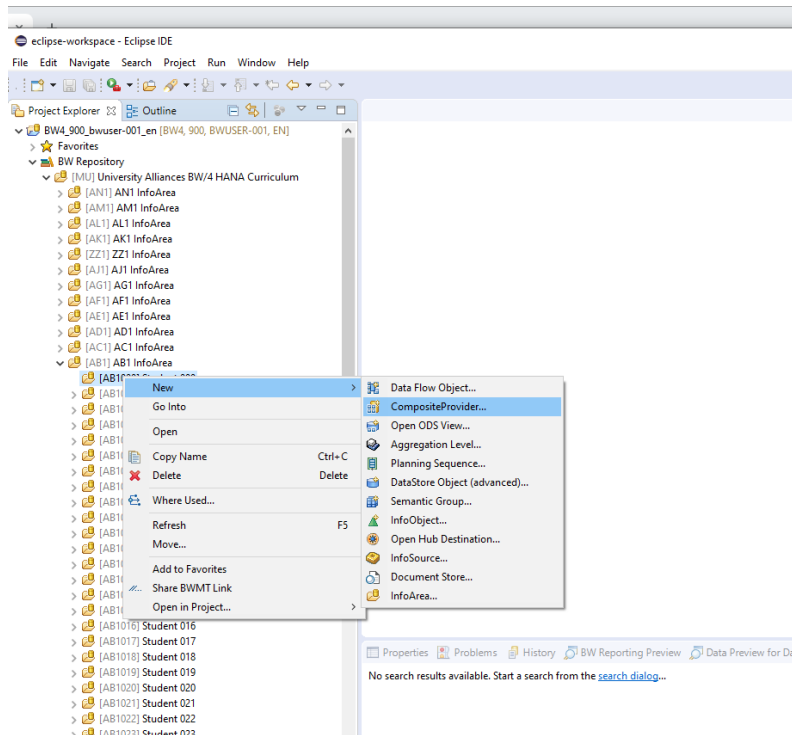


Figure 10

This is where you will build all the data structures for your data warehouse. This includes characteristics, key figures, aDSO, and composite provider.

- b. Close. Do not save.
- c. Now, go to the Master InfoArea **MU0** → **Composite Provider** → **MU0CP1 Sales Data Composite Provider**
- d. Double-click. (**Do not Edit or tamper with this data structure!**)

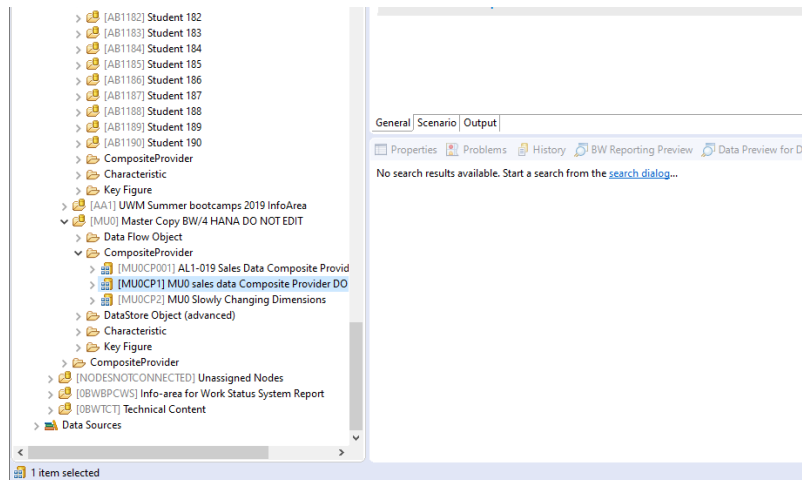


Figure 11

- e. Go to **Scenario** Tab
- f. Expand the Source objects as well as the Target objects.

**Question 4: Create a list of all key figures, then the dimensions. Within each dimension, list the characteristics.**

- g. Close the composite provider tab.

### 3. Advanced DataStore Object

The aDSO:

- Supports Field-based modeling and InfoObject-based modeling.
- Can serve as a source object for Open ODS views.
- Supports high-frequency data loads.
- Can contain up to 120 key fields.

- a. Open Eclipse and go to your own InfoArea provided by your instructor. Right click on your student **InfoArea** → **New** → **DataStore Object (advanced)**.

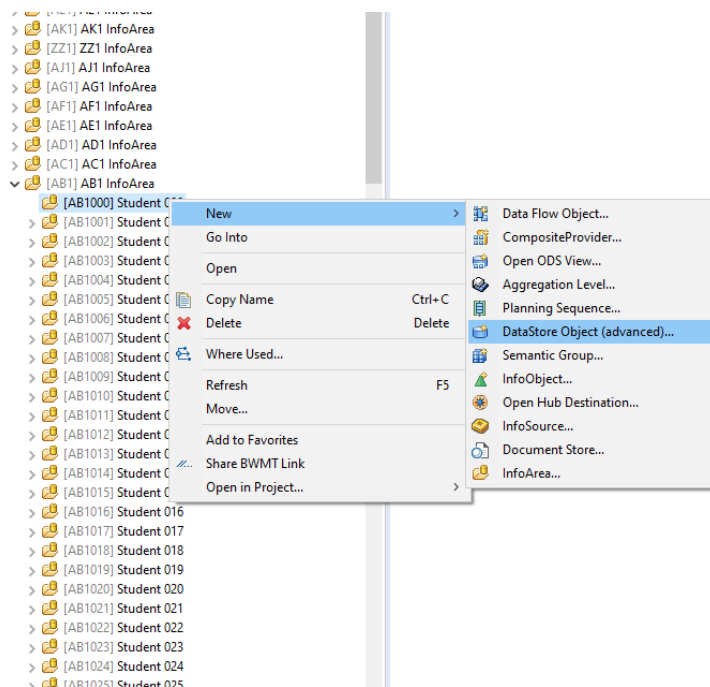


Figure 12

*Note: You can use two types of templates (from a DataSource or InfoProvider) or create a DataStore Object without a template.*

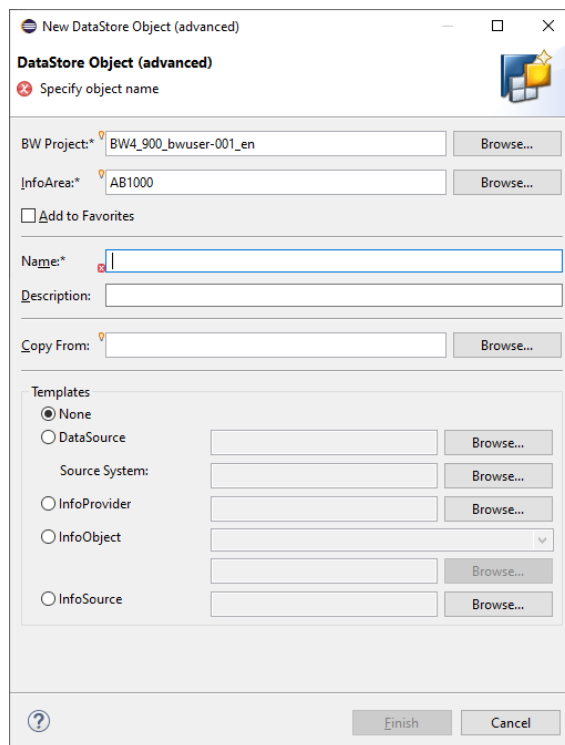


Figure 13

**b. Close** the window. Do not save.

You are now familiar with navigating in the SAP BW/4HANA 2.0 system using Eclipse SAP BW modeling tools. We will use this sort of navigation and naming convention to build the data warehouse in subsequent exercises. We will also populate this data warehouse using ETL techniques.

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